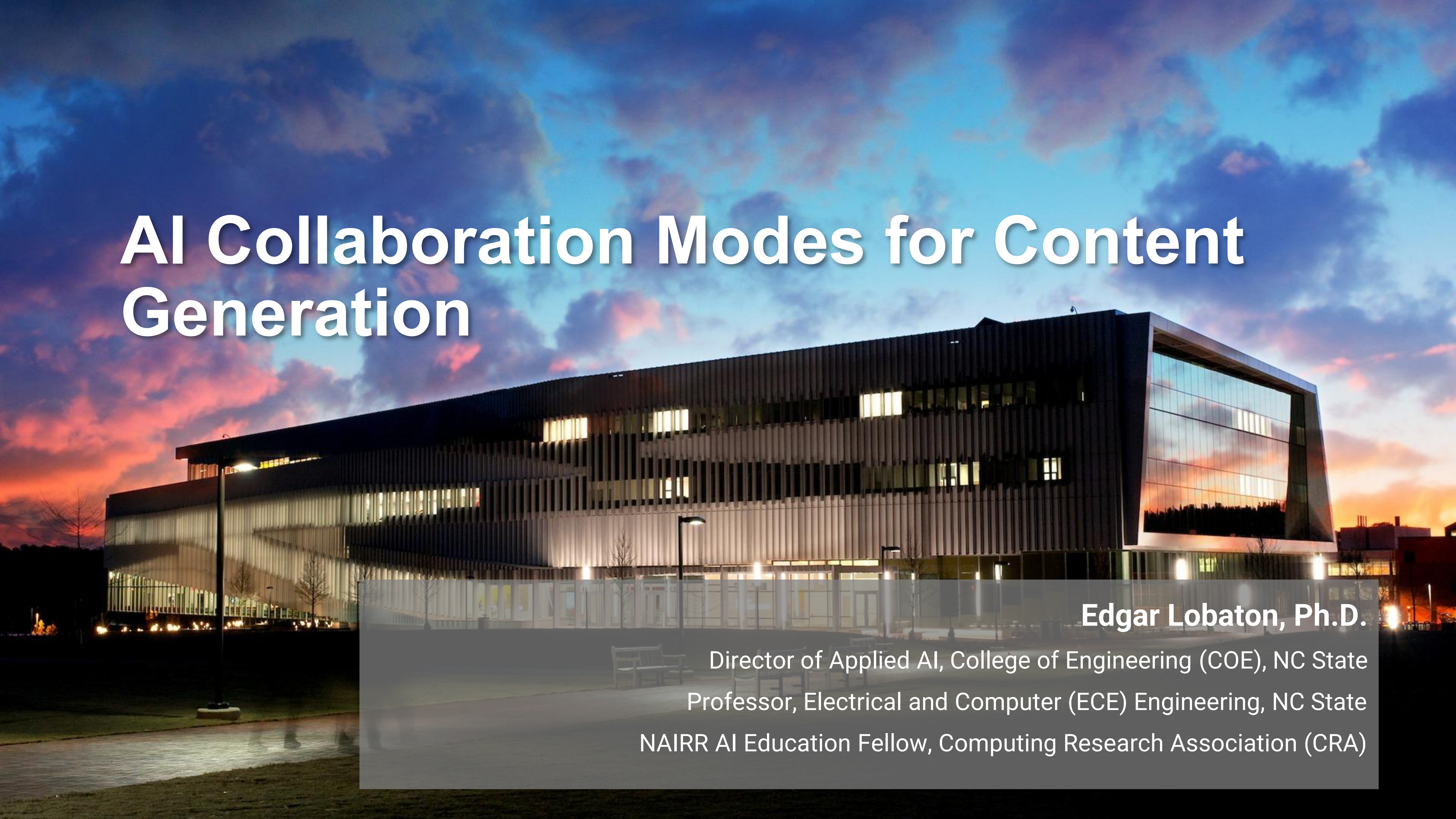


AI Collaboration Modes for Content Generation

Edgar Lobaton, Ph.D.

Director of Applied AI, College of Engineering (COE), NC State

Professor, Electrical and Computer (ECE) Engineering, NC State

NAIRR AI Education Fellow, Computing Research Association (CRA)

Three Modes of Working with an AI Code Assistant

NC STATE Engineering

Cyborgs, Centaurs, and Self-Automators — from a Harvard Business School study of AI-assisted knowledge work

Centaur

A clean division of labor. You decide what needs to happen, then delegate one bounded piece to the AI — and take control back to verify it.

Cyborg

Continuous interleaving, not a clean split. You and the AI go back and forth repeatedly within the same task — no clear line between “your part” and “its part.”

Self-Automator

The whole task goes to the AI. One high-level prompt, minimal ongoing engagement, review only at the end.

Let’s think of examples of use on non-coding tasks

https://www.hbs.edu/ris/Publication%20Files/26-036_e7d0e59a-904c-49f1-b610-56eb2bdfe6f9.pdf

NC STATE Engineering

Same ML Task, Three Different Workflows

What each mode looks like when building a model with an AI code assistant

	Centaur	Cyborg	Self-Automator
Your role	*Decide architecture & strategy yourself*	*Co-write live — start, edit, hand off, repeat*	*Write one high-level prompt*
AI's role	*Handles one bounded, delegated task*	*Continuously drafts, edits, explains errors with you*	*Runs the pipeline end-to-end, largely unattended*
Example prompt	*"Write the training loop for this architecture"*	*Live back-and-forth fixing a failing test together*	*"Build a model that predicts X from this dataset"*

None of these is "the right way" — the point is recognizing which mode you're in and choosing it on purpose.

The Scaffolding and the Engine

Bridging the Gap between CS Pedagogy and Engineering Application in the Agentic AI Era

Edgar Lobaton, Ph.D.

Director of Applied AI, College of Engineering (COE), NC State

Professor, Electrical and Computer (ECE) Engineering, NC State

NAIRR AI Education Fellow, Computing Research Association (CRA)

Code Generation at Scale

92%[1]

US-based developers who report actively using GenAI to accelerate tasks.

20–30%[2]

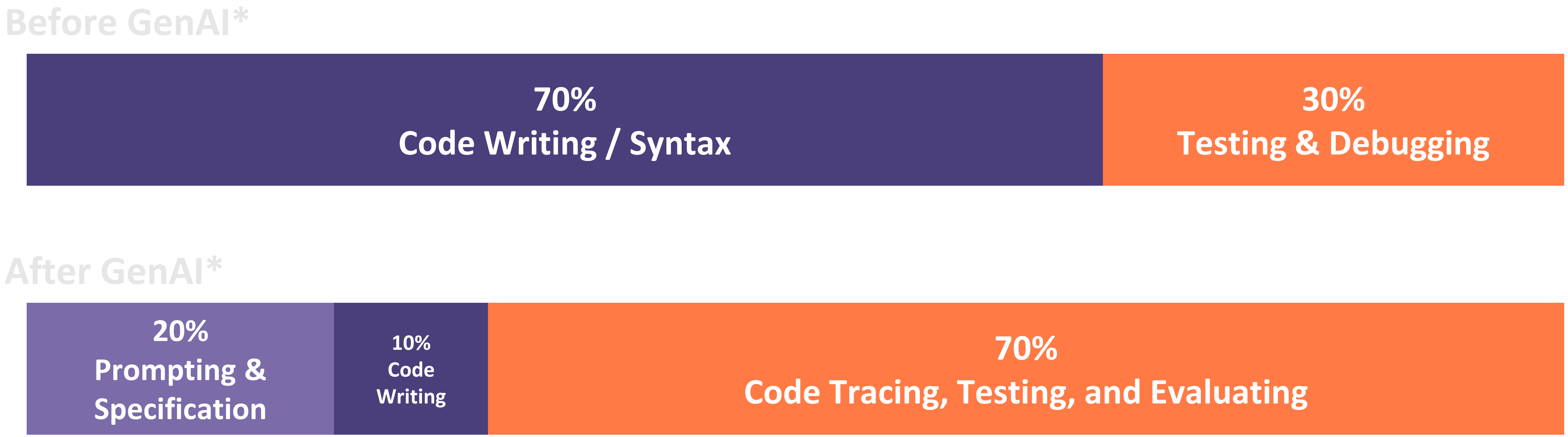
The volume of all new code currently written entirely by AI at companies like Google and Microsoft.

>95%[3]

The proportion of AI-generated code bases in a quarter of recent Y Combinator startups.

Code generation is rapidly becoming a solved problem. The value of human engineering is shifting from writing syntax to orchestrating systems.

The Shifting Skillset of the Modern Engr.



Engineering students must transition from being expert writers of code to expert evaluators of code.

*** These are estimates**

The Educational Dilemma of the Oracle

Step 1: Student inputs a zero-shot prompt (no context).

Step 2: Oracle AI outputs perfect, context-free code.

Step 3: Cognitive Bypass.

✗ ~~**Computational Thinking**~~

Behavioral Research Insights:

⚠ The Paradox

Students distrust AI accuracy yet prefer it for basic questions to avoid the social anxiety of asking human tutors.

⚠ The Result

Unrestricted access leads to AI acting as a direct answer engine, fostering superficial mimicry, overreliance, and an inability to independently troubleshoot.

CS Methodology: Imposing Guardrails

- Hidden system instructors sit between the student and the model, filtering very response

Student Prompt

↓

Hidden System Prompt Filter

↓

LLM

The Mechanism

Hidden system instructions force the AI to act as a Socratic tutor.

The Rule

Never give the direct answer. The AI is restricted to generating hints, clarifying errors, and forcing the student to write the actual code.

The Outcome

Approximates a 1:1 teacher-student ratio, preserving the cognitive friction necessary for foundational syntax mastery.

Identifying the Gap for Engineering

CS guardrails are design to teach how the loop works.

The Conflict

What if a graduate student already understands the underlying logic, but needs to process 10,000 rows of thermodynamic data?

The Gap

When coding is a secondary goal, artificial guardrails — specifically the Socratic withholding of code — become a hindrance. The engineer needs the answer to advance the science.

Divergent Paths in Academic AI

	Computer Science	Engineering
The Goal	Learning to program is the primary object.	Coding is a secondary tool used to solve a physical or mathematical problem.
The End Product	A software application or algorithm.	A scientific finding, a simulation, or a model.
AI's Ideal Role	Socratic Tutor (Forces productive struggle).	High-velocity Co-Pilot (Accelerates computation).
Primary Risk	Bypassing syntax mastery and logic building.	Hallucinated physics or mathematical inaccuracies in the final output.

AI as Rigorous Scientific Co-Pilot

The Zero-Shot Prompt

“Write a python script to model heat transfer.”

Result: High risk of hallucinated physics, generic outputs, and hidden logical errors.

The Engineering Prompt

“Using the SciPy library, establish a grid. Assume boundary conditions X and Y. Apply the specific heat capacity of titanium. Generate a simulation.”

Result: Grounded, domain-specific generation ready for evaluation.

Effective engineering prompts require deep domain expertise, precise specification, and rigorous problem decomposition.

Integrating the Engine into the Lab

NC STATE Engineering

1

Decompose

Students must break down the engineering problem into mathematical steps or pseudocode before touching the AI.

2

Generate

Remove artificial guardrails. Allow full use of GenAI to rapidly generate the required Python or MATLAB scripts based on the decomposition.

3

Verify

Shift grading and focus toward risk assessment and system evaluation. Does this code obey the laws of physics? Do the boundary conditions hold?

References

1. https://github.blog/news-insights/research/survey-reveals-ais-impact-on-the-developer-experience/
2. https://www.cnbc.com/2025/04/29/satya-nadella-says-as-much-as-30percent-of-microsoft-code-is-written-by-ai.html
3. https://techcrunch.com/2025/03/06/a-quarter-of-startups-in-ycs-current-cohort-have-codebases-that-are-almost-entirely-ai-generated/
4. H. Tomisu, J. Ueda, T. Yamanaka, “The cognitive mirror: a framework for Ai-powered metacognition and self-regulated learning,” Frontiers of Education, 2025.

Designing an ML Workflow

Edgar Lobaton, Ph.D.

Director of Applied AI, College of Engineering (COE), NC State

Professor, Electrical and Computer (ECE) Engineering, NC State

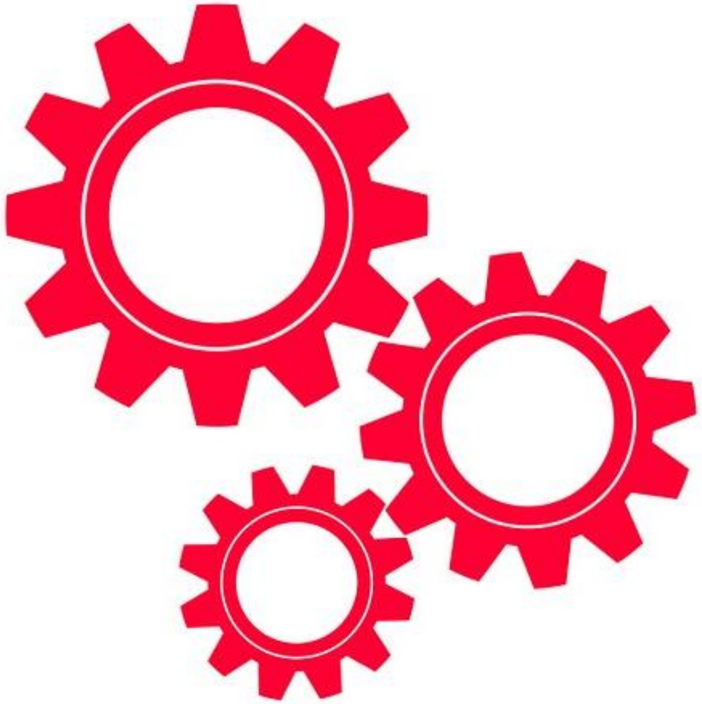
NAIRR AI Education Fellow, Computing Research Association (CRA)

The Problem: Synthetic Gait Classification

- We are aiming to develop a personalize model that determine when a person is walking fast or slow. We have conducted two recording sessions, in which the individual switches between fast and slow walking periodically. Let’s build a model that predicts which gait it is observed in the data.

The Problem: Synthetic Gait Classification

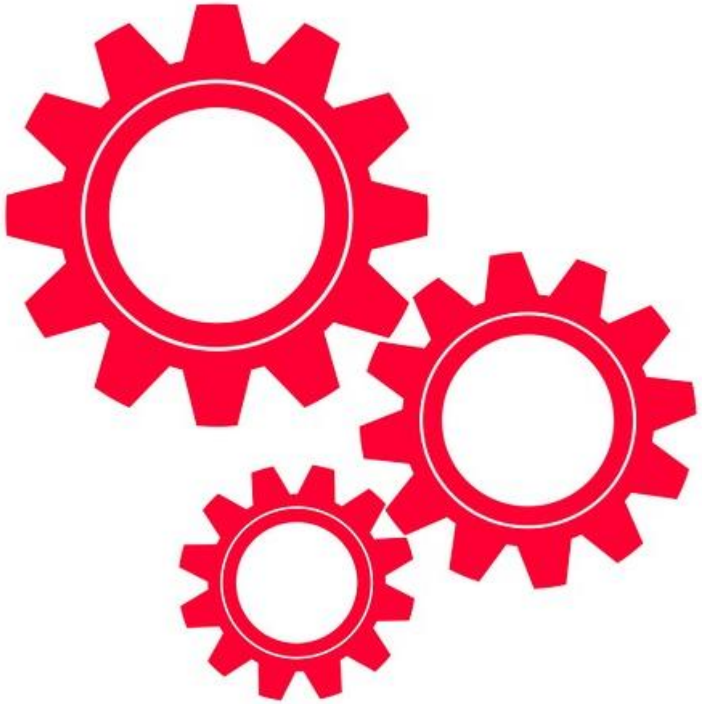
- Let’s do this as a Self-Automator



NC STATE Engineering

The Problem: Synthetic Gait Classification

- Let’s do this as a Centaur



The Problem: Synthetic Gait Classification

- Let’s refine as a Cyborg